

ENGR 2303: Electrical Science

HW 9_Ch9

1) Given the sinusoidal voltage $v(t) = 50 \cos(30t + 10^\circ)$ V, find:

(a) the amplitude V_m , (b) the period T , (c) the frequency f , and (d) $v(t)$ at $t = 10$ ms.

2) Given $v_1 = 20 \sin(\omega t + 60^\circ)$ and $v_2 = 60 \cos(\omega t - 10^\circ)$, determine the phase angle between the two sinusoids and which one lags the other.

3) Evaluate the following complex numbers and leave your results in polar form.

(a) $5 \angle 30^\circ \left(6 - j8 + \frac{3 \angle 60^\circ}{2 + j} \right)$

(b) $\frac{(10 \angle 60^\circ)(35 \angle -50^\circ)}{(2 + j6) - (5 + j)}$

4) Find the phasors corresponding to the following signals.

(a) $v(t) = 21 \cos(4t - 15^\circ)$ V

(b) $i(t) = -8 \sin(10t + 70^\circ)$ mA

(c) $v(t) = 120 \sin(10t - 50^\circ)$ V

(d) $i(t) = -60 \cos(30t + 10^\circ)$ mA

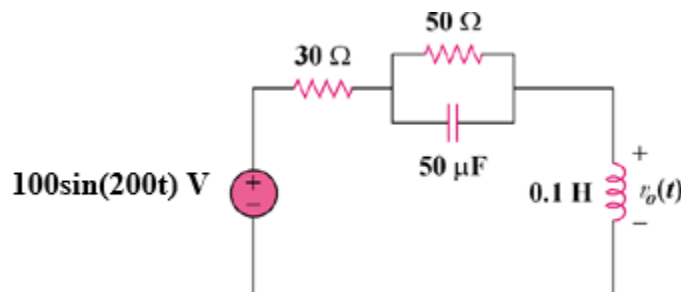
5) Evaluate the following complex numbers:

(a) $\frac{(2 + j3)}{1 - j6} + \frac{(7 - j8)}{-5 + j11}$

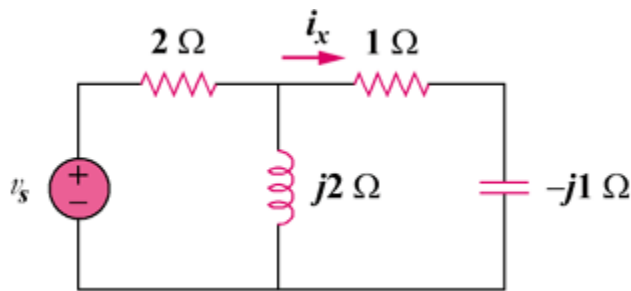
(b) $\frac{(5 \angle 10^\circ)(10 \angle -40^\circ)}{(4 \angle -80^\circ)(-6 \angle 50^\circ)}$

(c) A series RLC circuit has $R = 80 \Omega$, $L = 240$ mH, and $C = 5$ mF. If the input voltage is $v(t) = 10 \cos 20t$, find the current flowing through the circuit.

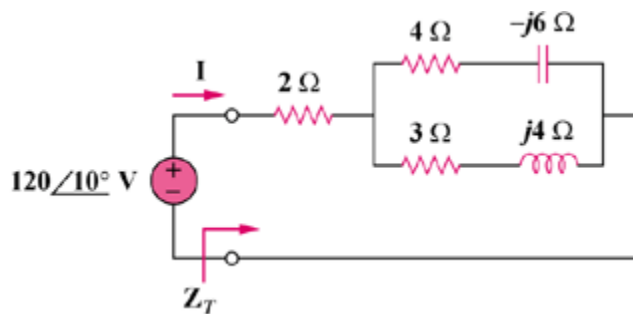
6) Calculate $v_o(t)$ in the circuit below.



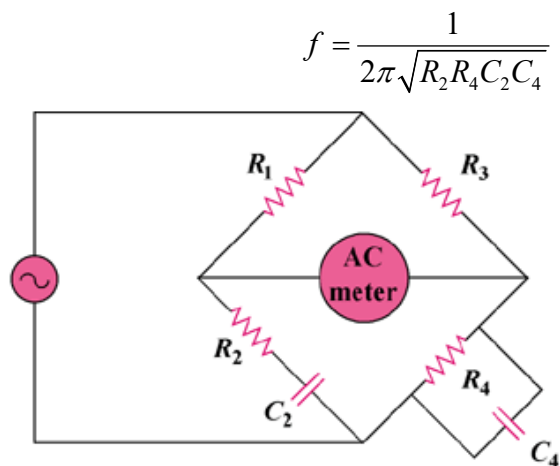
7) Find $v_s(t)$ in the circuit below if the current i_x through the $1\text{-}\Omega$ resistor is $0.5 \sin 200t$ A.



8) Determine $\mathbf{I}$ and $\mathbf{Z}_T$ in the circuit below.



9) The ac bridge circuit below is called a *Wien bridge*. It is used for measuring the frequency of a source. Show that when the bridge is balanced,



10) Figure below shows a parallel combination of an inductance and a resistance. If it is desired to connect a capacitor in series with the parallel combination such that the net impedance is resistive at 10 MHz, what is the required value of C ?

